

Universal Employee Verification & Multi-Company Attendance System

Comprehensive 5-tier architecture illustrating client modules, optical scanning engines, real-time WebSocket communication, relational PostgreSQL persistence, and universal barcode permission lifecycle.

Version: 2.4 Production • **Stack:** Vanilla JS / CSS • Node.js • Express • PostgreSQL • Prisma • Socket.IO

Tier 1: Frontend Client Presentation Layer (SPA)

Vanilla JS (ES6+) • CSS3/SCSS • Bootstrap 5.3

Landing & History Router

Module Gateway cards with HTML5 History API (`pushState` & `popstate`) for forward/backward browser navigation.

Verification & Pass Generator

JsBarcode (Code 128) & QRCode.js for instant dynamic badge rendering and verified demographic records.

Optical Attendance Kiosk

Html5Qrcode engine with webcam stream, laser target overlay, 3s visual cooldown, and foreign barcode rejection.

Lifetime Attendance Portal

Webcam scanner & code search for month-wise Present, Absent, and Permission logs across tenure.

RESTful HTTP Requests & Bidirectional WebSockets

Tier 2: Communication & Security Gateway Tier

REST Routing Gateway

Handles employee provisioning, offboarding, attendance check-in/out, and monthly aggregation requests.

Company Auth & Permission Guard

Validates active company sessions and denies revoked/inactive barcode credentials with 403 Forbidden.

WebSocket Event Dispatcher

Broadcasts `attendance:punch` events in real-time to the manager's live scan activity stream console.

Controller Invocation & Business Logic Execution

Tier 3: Core Application & Business Logic Services

Modular Backend Controllers • Aggregators

Identity Controller

Generates cryptographic barcode payload and maintains multi-employment relational history.

Lifecycle & Lock Engine

Processes "Enter Last Date", permanently locks end dates, and switches active status to false.

Attendance Punch Engine

Smart toggle logic for daily Check-In & Check-Out with shift duration calculation and debounce lock.

Month-Wise Aggregator

Groups punch records by `YYYY-MM`, computing verified Present, Absent, and Permission metrics.

Prisma ORM Queries & Relational Data Mapping

Tier 4: Relational Persistence Layer (PostgreSQL)

PostgreSQL 18 • Prisma ORM Client • ACID Compliant

persons Table

`id (UUID PK), name, email (UNIQUE), employee_code (UNIQUE), mobile_number, photo_url, status.`

person_details Table

`id (UUID PK), person_id (FK), company_name, start_date, end_date, is_active, monthly_salary, barcode_data.`

attendances Table

`id (UUID PK), person_id (FK), punch_date, check_in_time, check_out_time, status, work_mode, notes.`

Tier 5: End-to-End Universal Barcode Lifecycle Workflow

Multi-Company Career Flow

1

Onboarding

Barcode generated with `start_date` & `is_active=true`.

2

Daily Kiosk

Camera kiosk records Check-In/Out & live WebSocket stream.

3

Offboard / Lock

Admin enters *Last Date*. Permanently locked; status=inactive.

4

Re-Activation

Next company adds experience & reactivates same barcode.